Fit Passport

One body. One fit identity. Any store.

FINAL PROPOSAL · August 2026 *Xiangchen Kong · Alyssa Qi · Jenny Cao · Nicolas Wang Live at **https://fit-passport.vercel.app***

> This is the current and final proposal for Fit Passport. It supersedes `Fit-Passport-Proposal-Original.pdf` and `Fit-Passport-Detailed-Proposal.pdf` in this folder, which are retained as history — §4 records what changed between them and why.

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1. Startup overview

Fit Passport is a consumer-owned fit profile. You tell it once about the clothes that already fit you. It turns that into a portable record of how your body relates to real garments — then, when you paste any product link, it reads that page's size chart, tells you which size to buy, shows every signal behind the answer, and lowers its own confidence out loud when the signals disagree.

The profile belongs to the shopper, not the store. It works from a pasted URL with no retailer integration, and it sharpens every time the shopper records what they kept or returned.

What we are not building: a body scanner, an avatar, or a virtual try-on. Those are picture problems. Fit is a memory problem, and the memory is already hanging in the shopper's closet.

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2. Problem and origin

Sizes do not agree, and never have. A medium in one brand is a large in another. A 32 waist measures anywhere from 31 to 34 inches of actual cloth. A jacket that fits your shoulders swims at the chest. The label is a guess presented as a fact.

What a shopper actually does today — this is the real competition, so it is worth naming precisely:

- Opens the size chart and tries to recall their own measurements
- Scrolls reviews for *"runs small, size up"* from someone built similarly
- Recalls their size at a *different* brand and guesses the offset
- Orders two sizes intending to return one
- Abandons the purchase

Each is a workaround for the same missing thing: **nobody keeps a record of how this person's body relates to real garments.** The information exists — the shopper paid for it in returns, shipping labels, and things worn twice — and it is discarded at every new purchase. The costs compound: money, time, and confidence, each bad outcome making the next purchase harder. Retailers absorb reverse logistics, and returned garments are frequently landfilled rather than resold.

⚠ **On the numbers.** Chinese apparel return rates are well documented in `docs/design/china-sizing-research.md`: brand-store returns rose from **24% (2021) to 35% (H1 2024)**, livestream sales commonly around **80%**, with *"尺码不合适"* (unsuitable size) named among the leading drivers. The commonly quoted figure of roughly a quarter of US online apparel being returned is **not independently sourced in our own files**; it is treated here as directional, and verifying it against a primary source is an open item.

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3. Current solution

A working web application. Nothing to install.

The closet. The shopper adds garments they own, usually by pasting a product URL, and reports **which way each one misses** on a signed scale — *too tight · a bit snug · just right · a bit roomy · too loose*. One tap, with *"just right"* pre-selected because that is the answer roughly three-quarters of the time in both public fit datasets — so the common case costs nothing and only the informative answers require effort.

That signed direction is the decision the product turns on. A star rating cannot distinguish *"this is strangling me"* from *"this is hanging off me"*, and **those imply opposite recommendations.** Every size-recommendation system in the published literature models fit as a bipolar ordinal; a unipolar quality score discards the field's standard signal at the point of entry.

The size check. Paste any product link. The engine reads the page's size chart — structured data, on-page tables in either orientation, English or Chinese headers, inch-to-centimetre conversion, Chinese 号型 codes — ranks every available size with a per-signal reason, and draws the ease: the wearer's silhouette with the garment's outline around it, at true proportion, centimetre gap printed beside it.

Explainability is architectural. The engine is a transparent rule-and-score model over chest, waist and shoulder in centimetres. **A language model only extracts product data; it never picks a size.** A recommendation that cannot be interrogated cannot be trusted, and trust is the entire product. So when no real size chart can be read, the app says so and caps its confidence; when measurements point one way and the closet another, confidence falls **and the reason is stated** — *"by your measurements, XL; by the strongest overall evidence, M."*

Outcome learning. Keep/return/exchange outcomes feed a per-user, per-brand bias term. Brand bias also learns directly from the closet, so it works from the first few garments rather than requiring a purchase history almost nobody keeps.

Community — explicitly a hypothesis. A follow graph, outfit feed, Q&A anchored in real closets, daily boards and an earned-only badge ladder are built and running. They exist because a size engine is a tool you stop needing once you know your sizes, so retention must come from somewhere. **Whether that holds for this audience is unproven**, and §9 tests it rather than assumes it.

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4. What has changed since the original proposal

From a size tool to a consumer-owned fit identity. A calculator is a feature that can be cloned in a weekend; a profile the shopper owns and carries cannot. That reframes the competitive question from *"is our recommendation better?"* to *"who owns the profile?"* — which every incumbent answers with *"the retailer."*

From measurement-led to closet-grounded. Most people do not know their measurements, and asking is expensive. The closet is cheaper to collect and stronger as evidence: a garment that fits is a measurement someone already took, on a real body, in a real brand. Measurements are now one signal among several, not the spine.

From black-box AI to a transparent engine. Deliberate, and it cost us the easier-sounding pitch. A scoring model can be tested, explained and audited; a learned model over the data we have would be neither — and would ask shoppers to hand body data to something that cannot account for itself.

From building more to validating demand. The largest change, and the reason for this document. The product is further along than the evidence is, so feature work stops here.

Community and badges moved from features to hypotheses. Now built — but held as *claims about retention that have not been tested*, and explicitly downstream of the interviews rather than ahead of them.

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5. Target customer and alternatives

Beachhead: US online apparel shoppers, roughly 20–35, buying across multiple brands and burned by sizing. They already keep informal mental notes about which brands run small; we turn that into a real record. The sharpest wedge inside that group buys **across sizing systems** — US, EU, UK, JP, CN — where guesswork is worst, no existing tool helps, and scoring in centimetres pays off directly.

APPROACH	WHO	WHERE IT LEAVES AN OPENING
Purchase/return collaborative signal	True Fit, Fit Analytics	Lives inside one retailer's checkout; the profile dies there
Predicted body model from few inputs	Bold Metrics, EyeFitU, Sizebay	Retailer-integrated; the shopper never holds the profile
Photo / 3D body scan	3DLOOK	High friction, high sensitivity, retailer-side
Image-based virtual try-on	Google VTO, the diffusion-model wave	Transfers **appearance, not fit** — Google's own TryOnDiffusion states it does not promise fit

⚠ All vendor accuracy and returns-reduction figures are **self-reported**, several vendors' pages are internally inconsistent, and we obtained no third-party benchmark (`docs/design/fit-algorithm-research.md`). They are evidence the approach is commercially viable — **not** performance we can claim or match.

The alternatives that actually matter are not vendors. They are the five workarounds in §2: the size chart, the reviews, the memory, buying two sizes, and giving up. Those are free, familiar, and on every product page. **Any honest read of our position starts there**, because a shopper switching to us abandons a habit, not a subscription — which also sets the bar for the interviews: not *"would you use this?"* but *"what did you do last time, and what would have had to be true for you to do something else?"*

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6. Current status

The MVP works and is deployed at `fit-passport.vercel.app`, on Vercel with a Neon Postgres database. **265 automated tests pass.**

AREA	STATE
Transparent fit engine, per-signal reasoning, signed fit scale	Built
Product-page extraction (structured data, tables, chart-image OCR, 号型)	Built
Confidence that falls when signals disagree, with the reason stated	Built
Ease figure on the size check	Built
Closet, collections, edit history, outcome recording	Built
Brand bias learned from the closet and from outcomes	Built
Privacy: share a closet by code, never measurements	Built, enforced in the data model
Community: follow feed, Q&A with closet evidence, boards, badges	Built
Moderation: report, block, review queue, rate limiting	Built
Phone layout	Built

Two honest engineering limits. Some retailers refuse automated fetches outright — measured in production, H&M returns 403 — and we have declined to buy past that (§10). And item photos currently live inside Postgres, capping a cohort at roughly 340 users before writes fail for everyone; that is a code change plus a free storage tier, and it is required before inviting a real group.

> **The point that governs everything below: a finished product is not a validated market.** We have built the thing. We have not shown that anyone wants it badly enough to change what they do. Those are different claims, and conflating them is the most common way a project like this fails while looking healthy.

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7. Business model hypotheses

None of this is a plan. All of it is a hypothesis, stated with the condition that would have to hold.

H1 — Freemium with a premium tier. Core size check free; depth is paid — unlimited closet history, deeper analytics, advanced finishes and badges. *Holds only if* a meaningful minority use the product often enough for depth to matter. Contradicted if interviews show it is used a handful of times and abandoned once someone knows their sizes.

H2 — Affiliate on the purchase. When we give the right size and the shopper buys with confidence, an affiliate link aligns us with *the purchase working out* rather than with impressions. *Holds only if* we sit close enough to the buying moment to earn attribution — and only if it can be done without the recommendation becoming a sales pitch, which would destroy the trust the product is built on.

H3 — A fit API for retailers, later. The same engine attacks a store's largest returns cost, with per-brand fit-truth as the pitch. *Holds only if* the crowd fit-knowledge becomes deep enough to be worth buying — which is downstream of consumer adoption, not parallel to it. Not being pursued now.

Every path is downstream of one asset: structured, honest, consumer-owned fit data at scale. We build that first; monetisation follows the data. Charging before the data exists would mean selling a calculator.

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8. Where this goes beyond the beachhead

Three directions. **None of them is built**, and each is listed with what actually exists today so the distinction is not blurred.

Budget styling contests — the retention bet, and a new idea since the original proposal. A theme plus a spending cap; members submit a look with itemised prices; the community votes; winners take commemorative special-metal badges that cannot be bought. Three cadences are designed — **$100 weekly, $1,000 monthly, $10,000 seasonal** — because *a budget makes taste comparable and levels the field*, which is what turns a feed into a competition anyone can enter. **Status today: designed in `docs/design/community-ecosystem.md`, no code.** The deliberate first step is to **run one $100 contest entirely by hand** and see whether people enter, before any event tooling is written. Two open questions we would be testing: does a budget contest attract the fashion audience or only bargain hunters, and do prizes stay status-only or become real money — the latter changes the legal picture materially.

Pet fit — carried over from the original proposal, still a stretch. Harnesses and coats have the same problem in a worse form: sizing is wildly inconsistent, **the wearer cannot report discomfort**, and returns are awkward. **Status today: a `PetProfile` table exists in the schema — species, breed, neck, chest girth, back length, known-good gear — and no application code reads or writes it.** A placeholder, not a feature. It stays parked until the human product is validated; building a second unvalidated audience would be the same mistake twice.

Non-apparel categories. The engine scores measurements, not garment types, so nothing in the architecture is specific to shirts — shoes and rings are the same shape of problem. Long-range, and mentioned only because it is why the engine was built on centimetres rather than size labels.

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9. Validation plan

Target: 20–25 shopper interviews, in two rounds so the second is reshaped by the first.

Round 1 — 8 to 10 interviews. Thirty minutes, semi-structured, **no product shown in the first half.** That half is about behaviour, not opinion: walk me through the last time you weren't sure about a size; what did you look at; what did you actually do; have you ordered two sizes, and what made you. Stated preference about a product someone has just been shown is close to worthless — we need what they did last month.

Five reads per interview:

1. **Problem intensity** — irritation or real cost? Target: ≥60% describe a specific recent instance unprompted.
2. **Current behaviour** — which of the five workarounds, and what does it cost them?
3. **Concept acceptance** — does closet-as-evidence land, or read as more work?
4. **Privacy** — how do they react to entering body data, and does *"measurements never leave; only your closet is shareable"* change that?
5. **Willingness to pay** — against the specific hypotheses in §7, never as an abstract "would you pay".

Round 2 — the remaining interviews, guide revised on Round 1's findings. Personas are built after this round, so they come from evidence.

Then usability testing, 8–10 sessions on the live app, measuring the closet-entry burden specifically: how many garments does someone add before stopping, and does the one-tap fit scale hold up outside our own assumptions about it?

Then a small beta, 10–15 users: recommendation agreement against sizes people actually own, repeat use within seven days, and whether recorded outcomes visibly improve later recommendations.

What would falsify the concept, written down in advance so it cannot be rationalised afterwards: if most interviewees describe sizing as a mild annoyance rather than a cost, if closet entry proves to be the barrier, or if honest uncertainty reads as weakness rather than trustworthiness — then the consumer product is the wrong wedge and the retailer-facing engine is the surviving idea.

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10. Risks and immediate next steps

Insufficient customer need. The central risk, and the reason for §9. The product's completeness makes it *easier* to overlook, not harder. Mitigation: behaviour-first interviews, and falsification conditions written before the data arrives.

Data-entry burden. The engine is only as good as the closet, and every field is a reason to quit. Hence one pre-selected tap instead of a dropdown, no free-text notes or prompted photos, and deliberately **no drag slider** — measured break-off on sliders reaches 37% on smartphones against 2.3% for discrete options. Mitigation: measure drop-off directly in usability testing rather than trusting the argument.

Open decision — the numeric input mode. A finer-grained alternative to the five-option tap was requested as a **1–20 comfort scale**. That form is *unipolar*, reproducing the exact defect the signed scale removes: 20 is comfortable, but 1 cannot say whether the garment is strangling the wearer or hanging off them. What shipped instead is the **same signed −10…+10 range at finer resolution**, tapped on a line with no marker at rest, so "not answered" stays distinguishable from "answered zero". **The outstanding decision is whether that substitution is accepted**, not whether to build it.

Extraction fragility. Retailer pages change, and some refuse automated requests. We have **declined stealth-proxy scraping** — not on price but on posture: the case law protecting public scraping expects technical access controls to be respected, and a 403 is one. Trading a governance story for size charts is a bad trade for a product whose differentiator is trustworthiness. The intended answer is a **browser extension** reading the page in the shopper's own browser, where they are already welcome — deliberately unbuilt until the interviews justify it.

Body-data privacy. Precise measurements are never exposed. Sharing by account code reveals brands, sizes and a coarse body type; the centimetre fields are not filtered out of the response, they are never read from the database at all. Every field is optional, a deactivated account disappears everywhere, and export and deletion are supported.

Open decision — what a shared code reveals. Whether the signed fit direction should be visible to a code-holder. It is closet information exactly like the already-public fit rating, and more useful. But direction **points at the body in a way a star rating does not**: enough signed reports read against published size charts form a system of inequalities about someone's measurements, and *"precise centimetres never leave"* is the hardest promise this product makes. Widening what a bearer code reveals is a governance decision, not a side effect — so it is parked rather than defaulted into.

Platform terms. The current hosting plan forbids commercial use — a licence term, not a resource limit, applying the moment the project charges anyone.

Immediate next steps

1. **Revise the interview guide** for a product that now exists — the current draft predates deployment and still asks hypothetical questions.
2. **Run Round 1** (8–10 interviews) and synthesise.
3. **Move item photos out of the database**, before any cohort beyond a handful.
4. **Revise the guide, run Round 2**, and build personas from the findings.
5. **Decide the browser extension on the evidence**, not before it.

Nothing on this list is a new feature. The next thing this project needs is not more product.

Supporting material: `docs/prospectus/` · `docs/business/` · `docs/design/` *(fit research, closet-signal and interaction-cost analysis, fetch strategy, cost model, identity threat model)* · `DEVLOG.md` · *the source in* `app-web/`.